Study designs

Biostatistics and Scientific Research

9-3-2026

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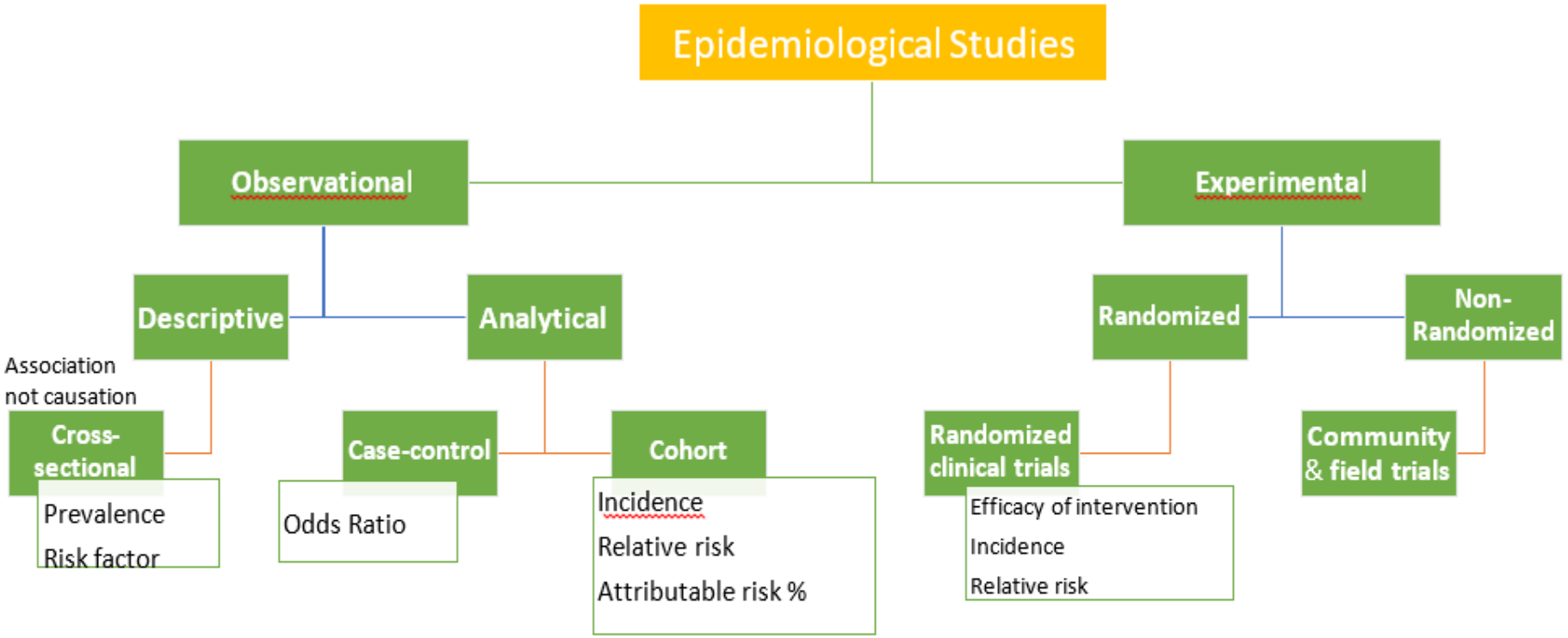
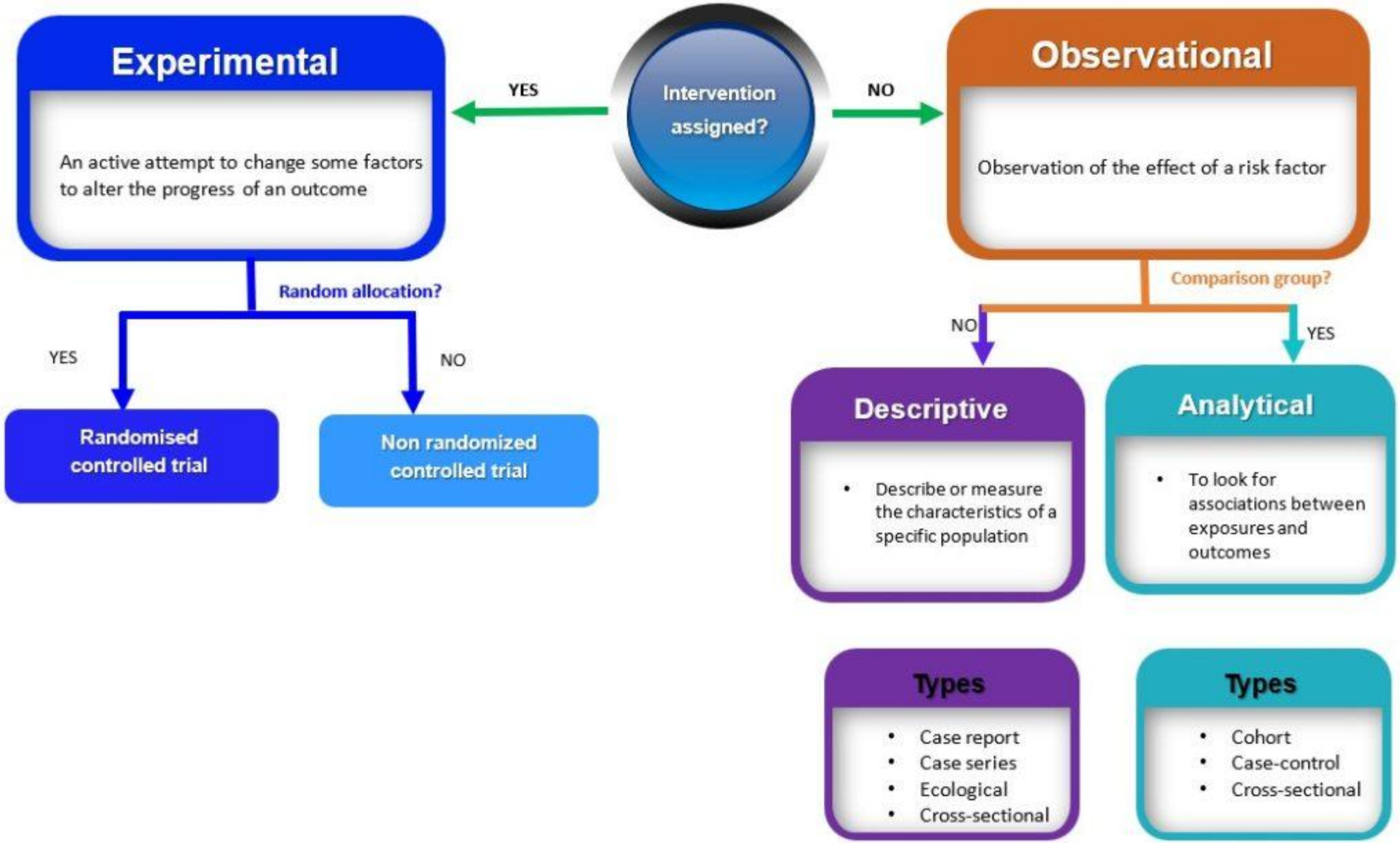
Second semester- Third year

Week 3

Study Design



Types of Descriptive Studies



Types of studies

Observational studies:

They are research methods that involve observing and recording the effect of a risk factor, diagnostic test, treatment without trying to change who is or isn't exposed to it.

Experimental studies:

They are ones where researchers introduce an intervention and study the effects.

Types of Observational studies

1. Descriptive studies:

- It is often the first step in an epidemiological investigation.
- It focuses on a description of the occurrence of a disease in a population.
- It is based on routinely available data or on data obtained in special surveys.
- The link between the exposure and the effect not analyzed.
- Usually based on mortality statistics .
- May examine pattern of death based on age, sex, ethnicity during period of time or in different countries

Types of Observational studies

2. Analytical studies:

- They go further by analyzing relationships between health status and other variables.
- They explore relationships and potential causes.
- They determine whether a relationship exists between variables
- They assess the strength and direction of that relationship.

Types of observational Studies

1. Cross sectional study:

- Provides information about a health condition (disease) that exist at a given time/during a given period.
- Data is collected simultaneously for all participants, through survey, questionnaire, or observation.
- It is called a snapshot of a population.
- It can be analytical.
- Quick and inexpensive.
- **Example:**

Prevalence of Hypertension and Associated Lifestyle Factors

Types of observational Studies

2. Case control study:

- It is used to identify and evaluate the association between exposures (risk factors) and outcomes (diseases or conditions).

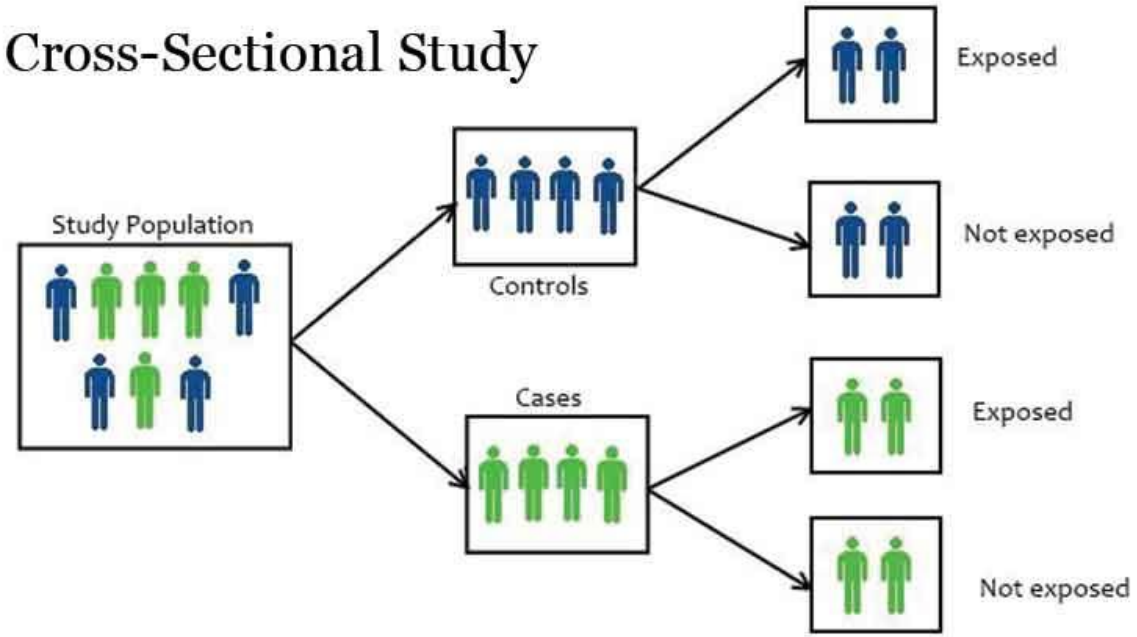
This design is particularly useful for studying rare diseases or conditions.

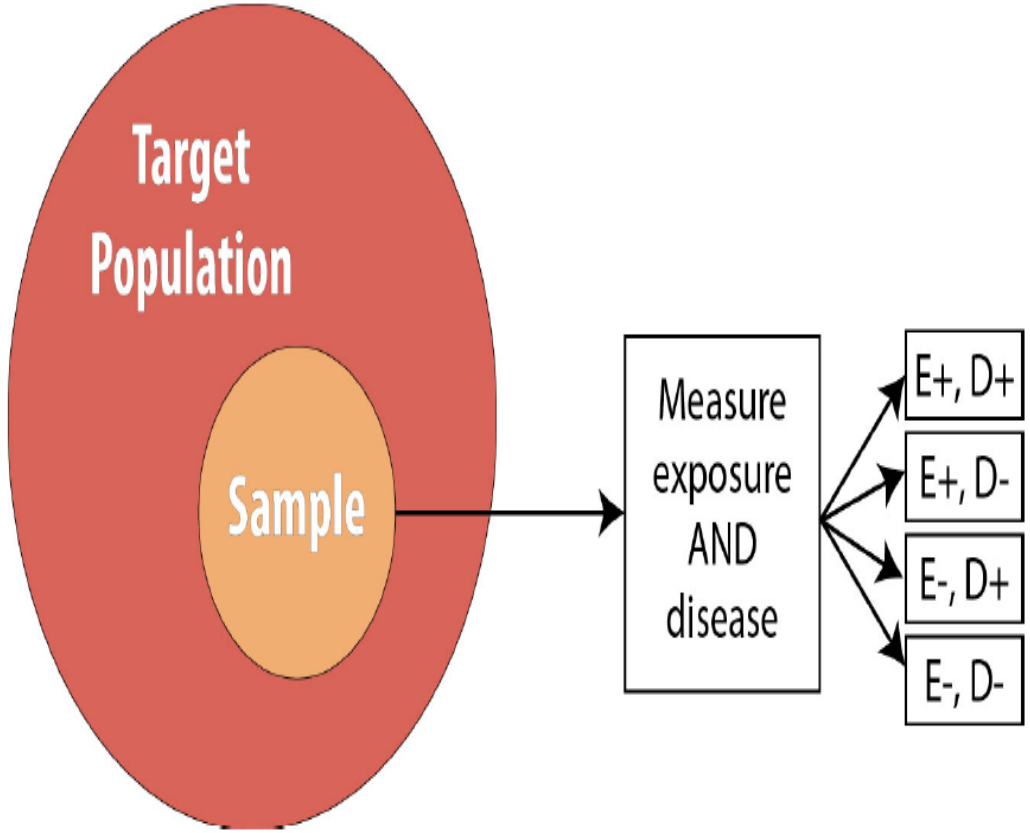
Example:

- Lung Cancer and Smoking
- Heart Disease and Diet
- The design is quick, inexpensive relatively simple, except that it is backward-looking (retrospective) based on the exposure histories of cases and controls.

Types of observational Studies

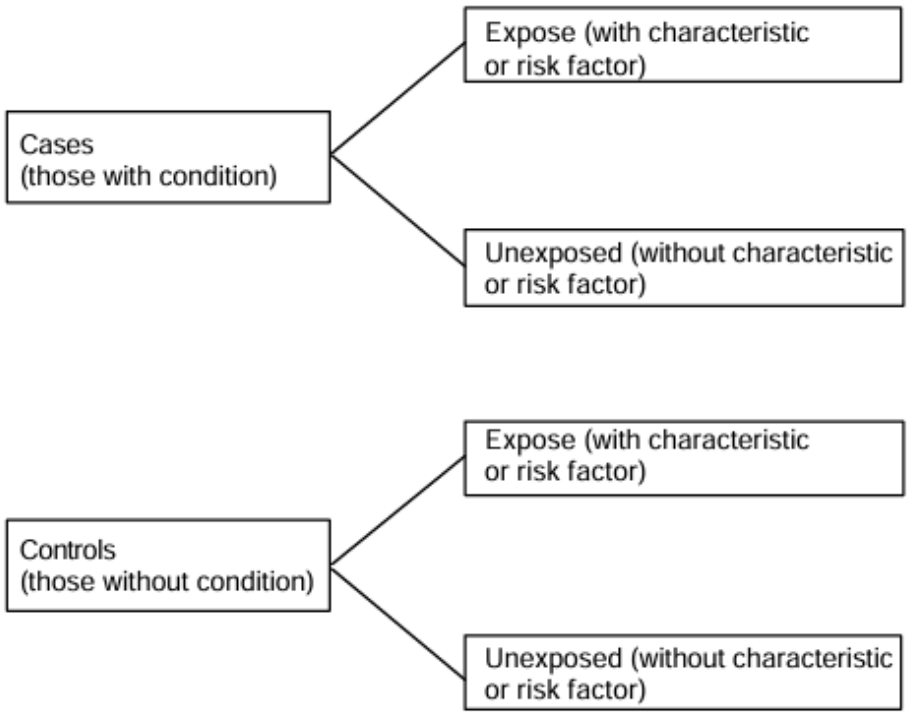
1. Cross sectional study design

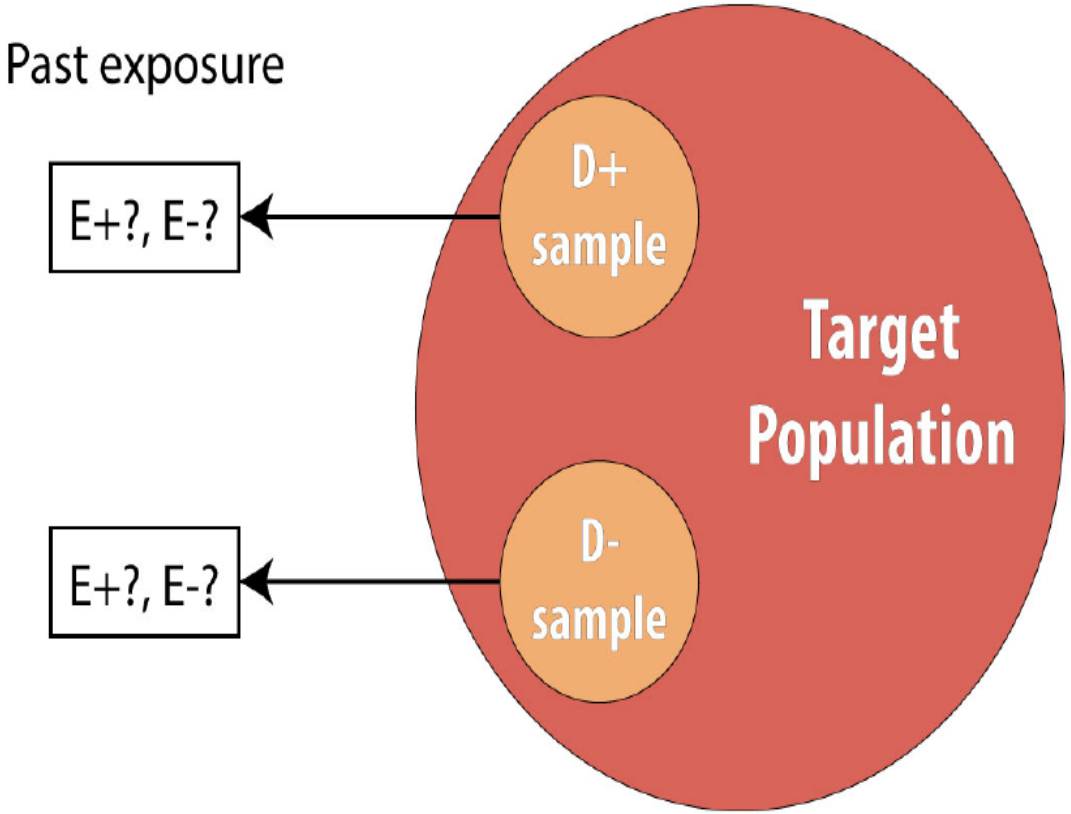




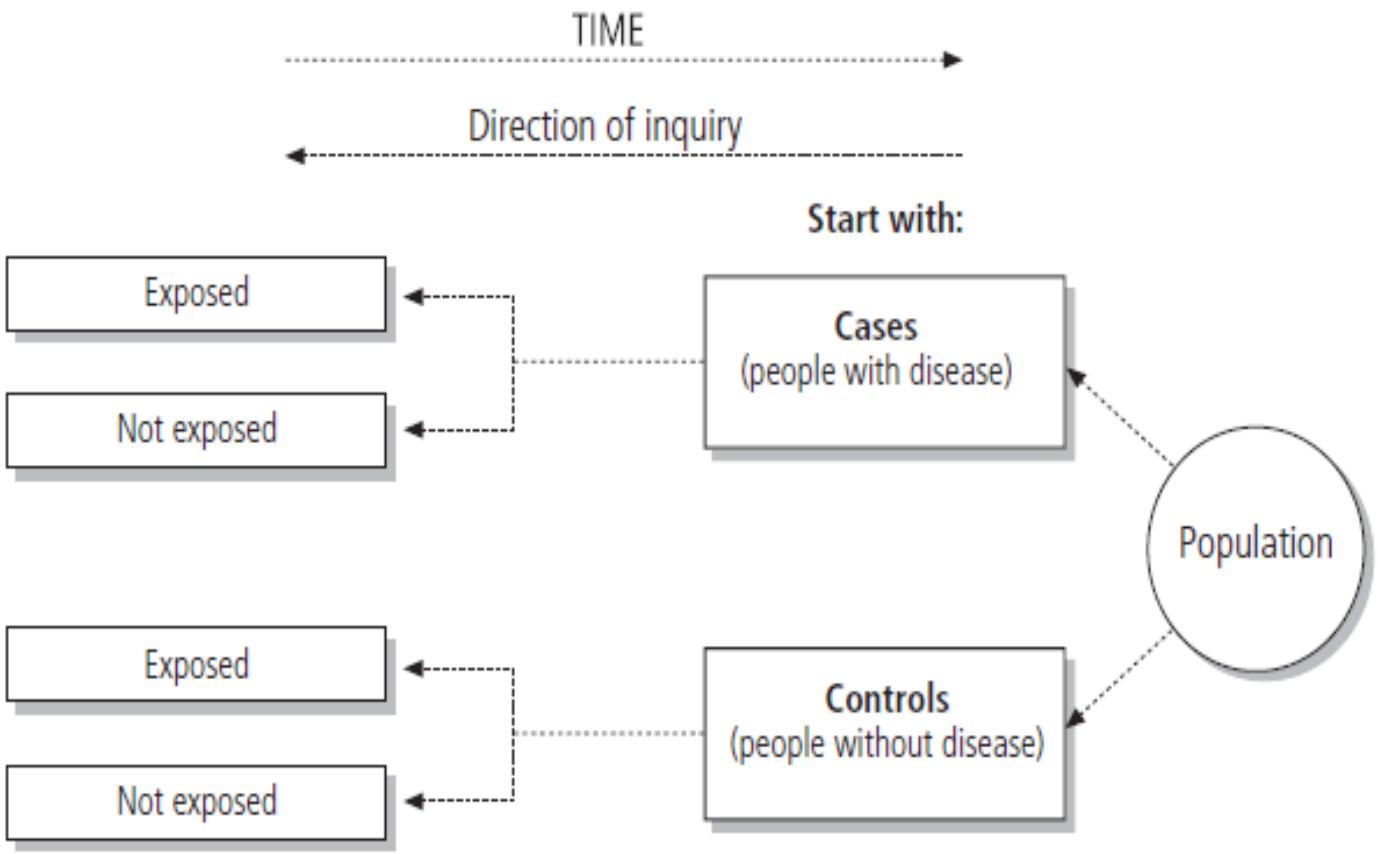
Types of observational Studies

2. Case control study:



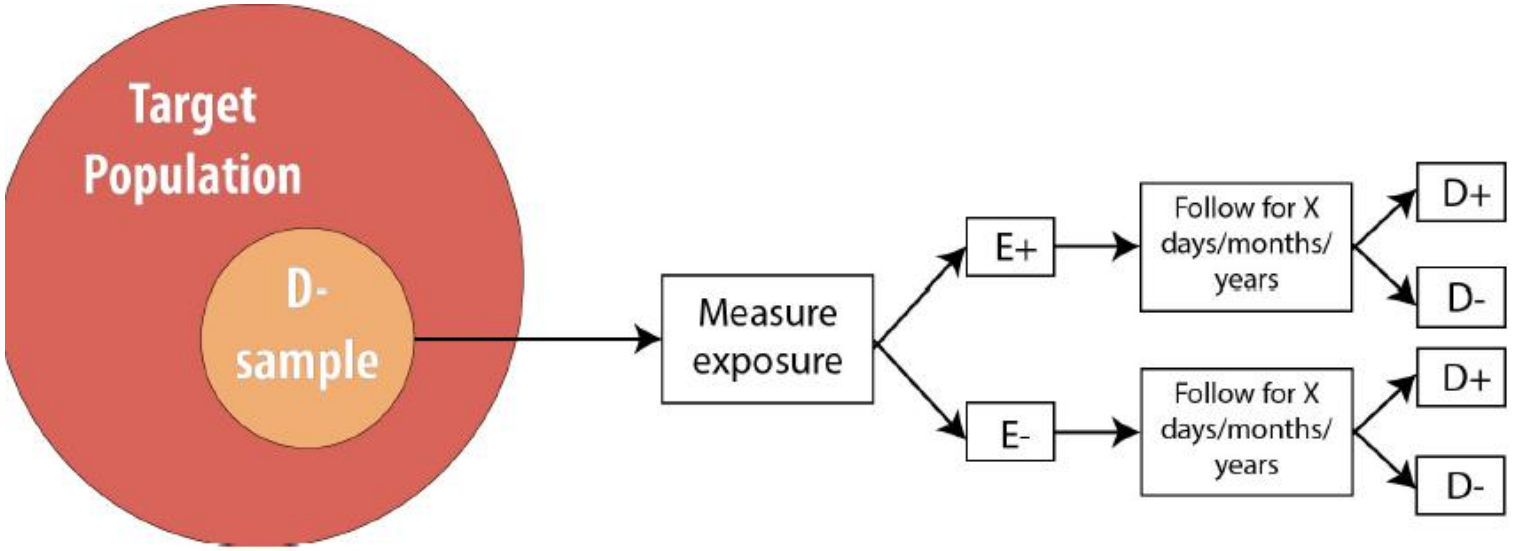


Case Control Study Design



Types of observational Studies

3. Cohort study:



Types of observational Studies

3. Cohort study:

- A cohort is a group of individuals who share a common characters within a specific time period, as being born in the same year.
- It is used to investigate the relationships between exposures (such as lifestyle factors, environmental influences, drugs) and outcomes (e.g. diseases) over time.
- Participants are typically classified based on their exposure status.

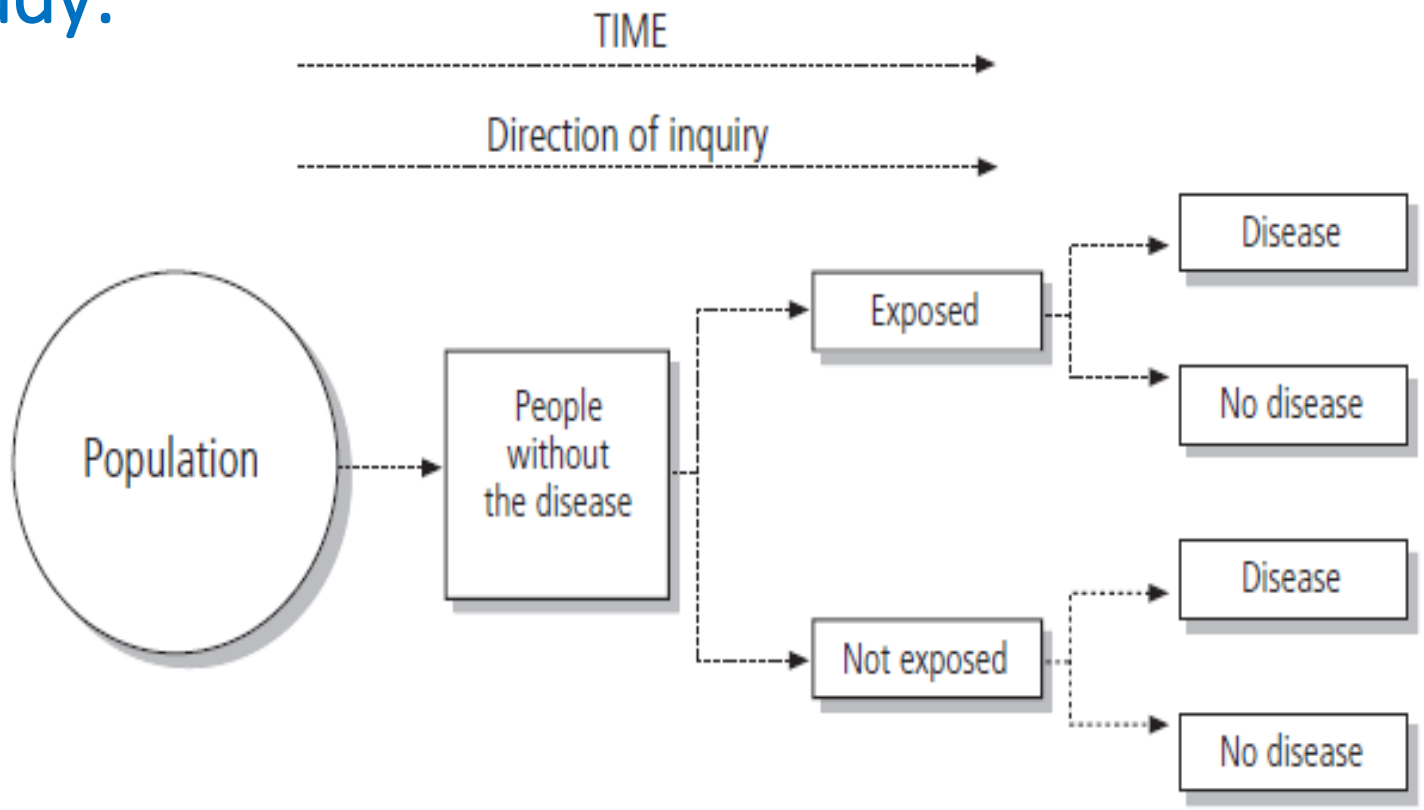
Types of observational Studies

3. Cohort study:

- Cohort studies follow participants over time, collecting data at multiple points to observe how exposures affect outcomes as they develop.
- Cohort studies provide the best information about the causation of disease and the most direct measurement of the risk of developing disease.
- It may require long periods.
- Example:
- Longitudinal Study on Smoking and Lung Cancer

Types of observational Studies

3. Cohort study:



Experimental study

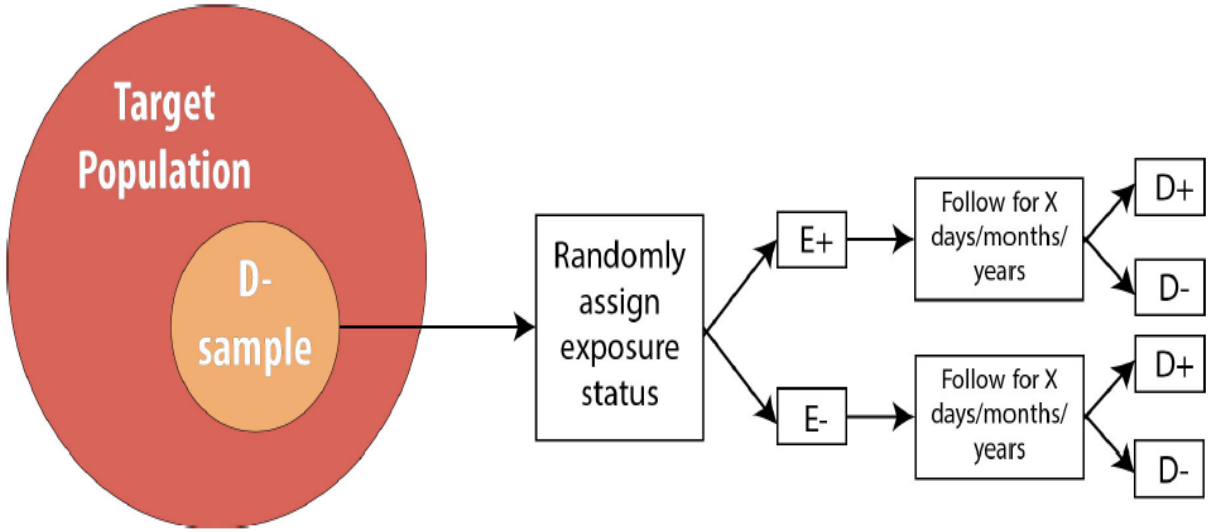
- Experimental studies are ones where researchers introduce an intervention and study the effects.
- Intervention involves attempting to change a variable in one or more groups of people.
- It is usually randomized, meaning the subjects are grouped by chance.
- The effects of an intervention are measured by comparing the outcome in the experimental group with that in a control group.
- An interventional study is usually designed as a randomized controlled trial, a field trial, or a community trial.

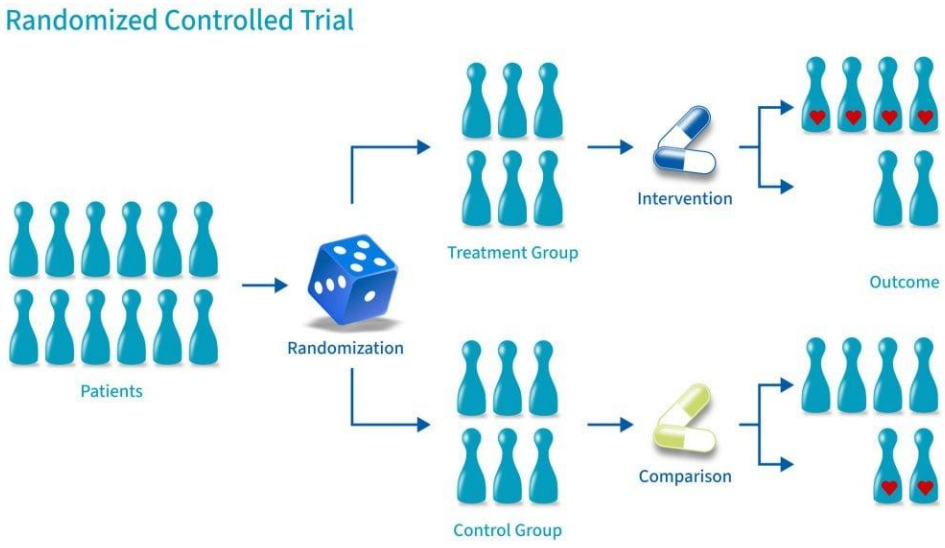
Randomized controlled trials study design

- A randomized control trial (RCT) is a type of study design that involves randomly assigning participants to either an experimental group or a control group to measure the effectiveness of an intervention or treatment.

Examples:

- Clinical Drug Trial
- Vaccine Efficacy Study





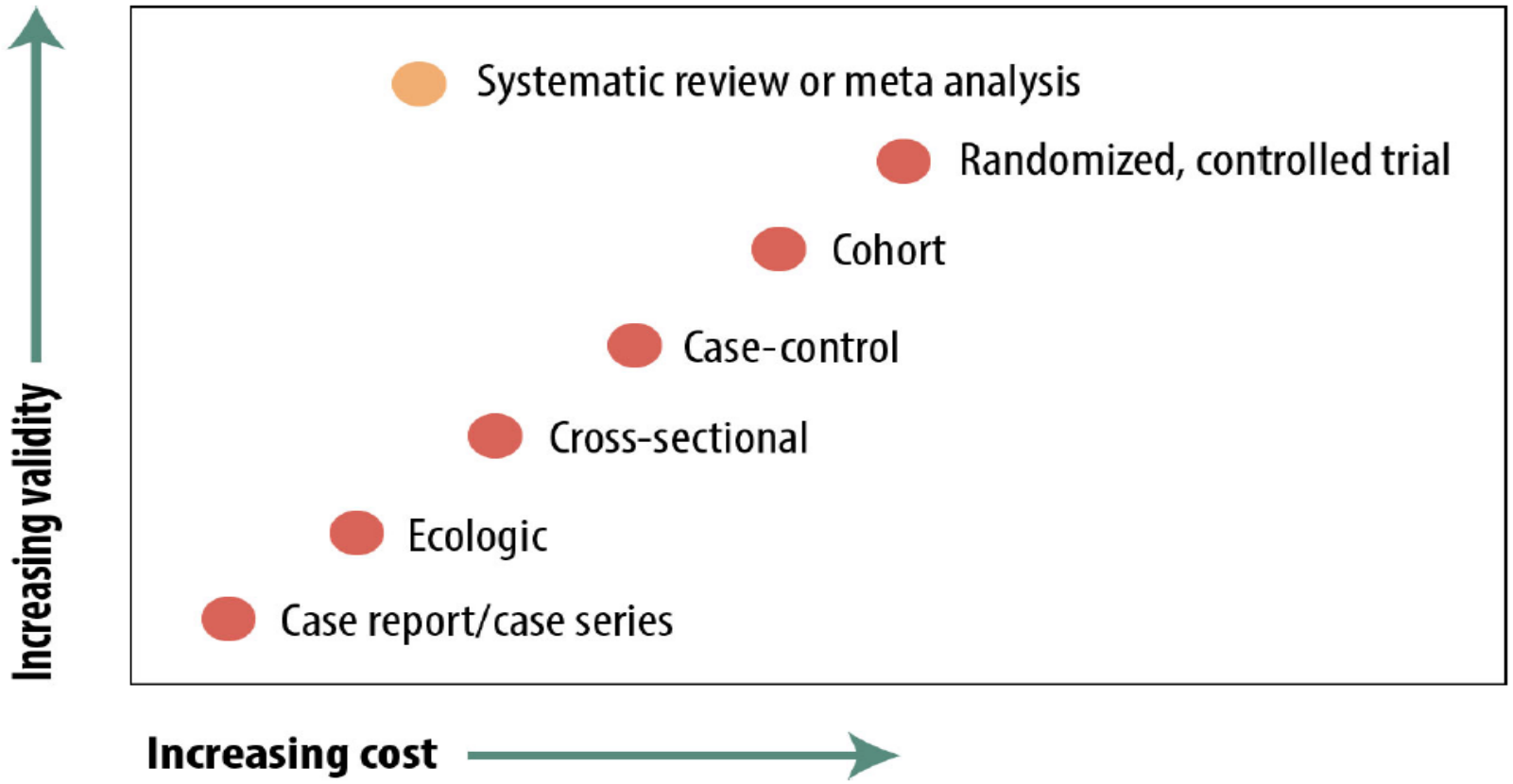

Systematic review or meta analysis
Randomized, controlled trial
Cohort
Case-control
Cross-sectional
Ecologic
Case report/case series
Increasing validity
Increasing cost

Figure 9-10

THANK
YOU